

Activity 5: Data Visualization Techniques

Complete Answers — R Programming (ggplot2, igraph, base R graphics)

This document contains complete R code solutions, generated visualizations, and interpretations for all five parts of Activity 5: Geospatial Data Visualization, Network Visualization, Temporal Data Representation, Multivariate Data Visualization, and an Integrated Data Visualization Exercise. Each answer includes ready-to-run R code followed by the resulting plot and a short interpretation.

1. Geospatial Data Visualization

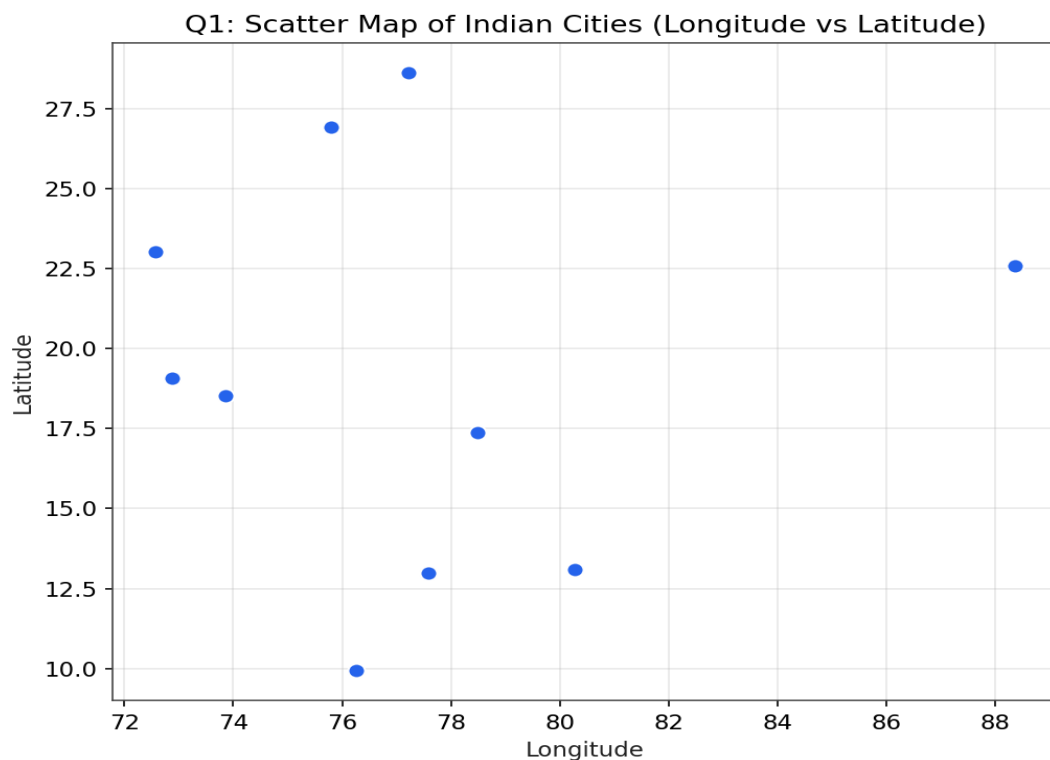
Dataset & Setup

```
library(ggplot2)

cities <- data.frame(
  City = c("Chennai", "Bangalore", "Hyderabad", "Mumbai", "Delhi",
           "Pune", "Kolkata", "Ahmedabad", "Jaipur", "Kochi"),
  Latitude = c(13.08, 12.97, 17.38, 19.07, 28.61, 18.52, 22.57, 23.02, 26.91, 9.93),
  Longitude = c(80.27, 77.59, 78.48, 72.87, 77.21, 73.85, 88.36, 72.57, 75.79, 76.26),
  Population = c(11, 13, 10, 20, 32, 7, 15, 8, 4, 2)
)
```

Q1. Scatter map showing all cities (longitude vs latitude)

```
ggplot(cities, aes(x = Longitude, y = Latitude)) +
  geom_point(color = "steelblue", size = 3) +
  labs(title = "Scatter Map of Indian Cities",
       x = "Longitude", y = "Latitude") +
  theme_minimal()
```



Output — Q1: Scatter map of city locations

Q2. Bubble map — bubble size represents population

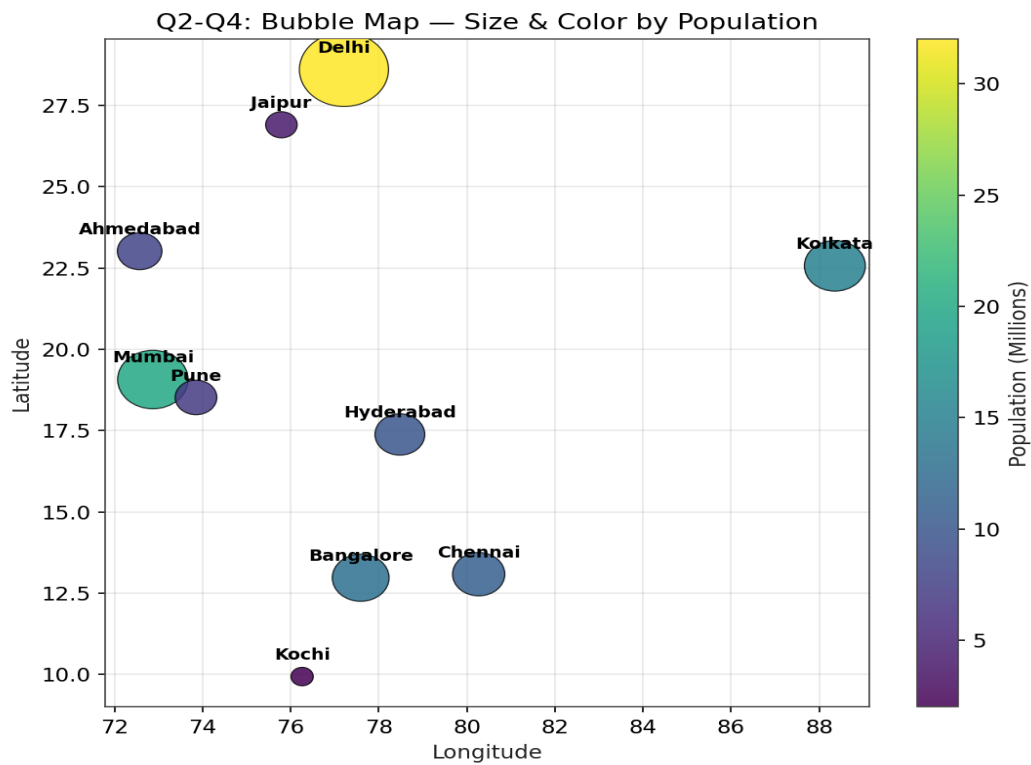
```
ggplot(cities, aes(x = Longitude, y = Latitude, size = Population)) +
  geom_point(color = "darkorange", alpha = 0.7) +
  scale_size(range = c(3, 15)) +
  labs(title = "Bubble Map of Population by City") +
  theme_minimal()
```

Q3. Label each city using geom_text()

```
ggplot(cities, aes(x = Longitude, y = Latitude, size = Population, color = Population)) +
  geom_point(alpha = 0.8) +
  geom_text(aes(label = City), vjust = -1.2, size = 3.5, color = "black") +
  scale_size(range = c(3, 15)) +
  scale_color_viridis_c() +
  labs(title = "Bubble Map with City Labels") +
  theme_minimal()
```

Q4. Different colors based on population size

The **color** aesthetic (`color = Population`) maps population to a continuous color scale (`scale_color_viridis_c()`), so cities with a larger population appear in a different shade/color than smaller ones — combined here with Q2 and Q3 in a single plot:



Output — Q2, Q3 & Q4 combined: Bubble map, labeled, colored by population

Q5. City with the highest population

Answer: From the plot, **Delhi** has the largest and most intensely colored bubble, corresponding to a population of **32 million** — the highest among the 10 cities. Mumbai (20M) and Kolkata (15M) follow as the second and third largest.

2. Network Visualization

Dataset & Setup

Note: The source table has a formatting issue where two rows ('B→D' and 'C→D') were merged into one line (B DC D). Based on context this has been read as two edges: **B → D** and **C → D**.

```
library(igraph)

edges <- data.frame(
  From = c("A", "A", "B", "C", "C", "D", "E", "F", "G", "H"),
  To   = c("B", "C", "D", "D", "E", "E", "F", "G", "H", "A")
)

g <- graph_from_data_frame(edges, directed = TRUE)
```

Q1 & Q2. Create the network graph; display labels and customize node colors

```
V(g)$color <- rainbow(vcount(g))

plot(g,
  vertex.label = V(g)$name,
  vertex.color = V(g)$color,
  vertex.size = 30,
  edge.arrow.size = 0.5,
  main = "Network Graph of Nodes A-H")
```

Q3. Degree of each node

```
degree(g, mode = "total")
```

Node	Degree
A	3
B	2
C	3
D	3
E	3
F	2
G	2
H	2

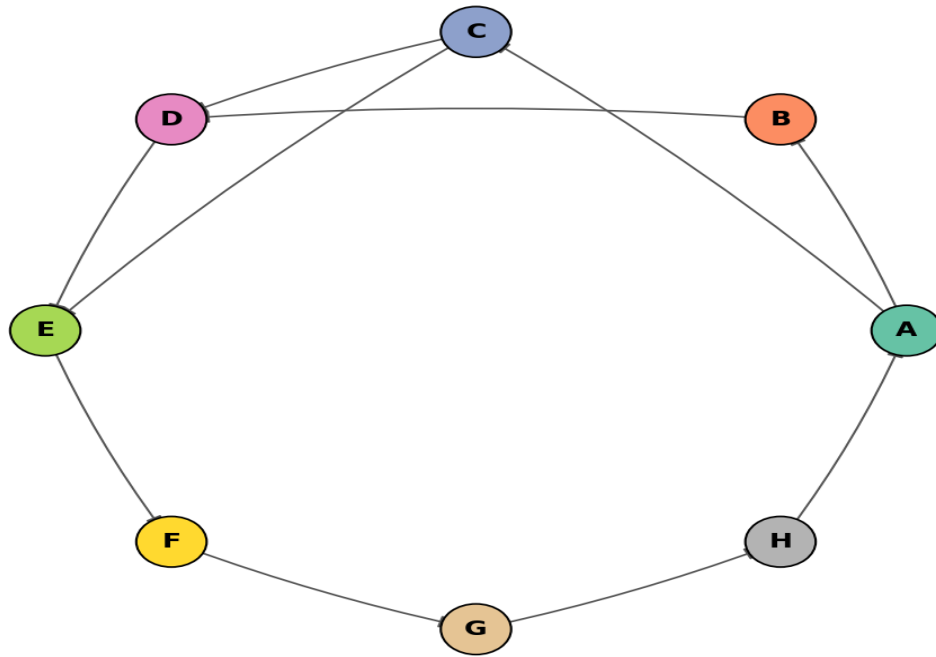
Q4. Node with the highest number of connections

Answer: Nodes **A, C, D, and E** are tied for the highest degree, each with **3** connections (A connects to B, C, and H; C connects to A, D, and E; D connects to B, C, and E; E connects to C, D, and F). All remaining nodes (B, F, G, H) have degree 2.

Q5. Plot the network using a circular layout

```
plot(g,
  layout = layout_in_circle(g),
  vertex.label = V(g)$name,
  vertex.color = V(g)$color,
  vertex.size = 30,
  edge.arrow.size = 0.5,
  main = "Network Graph - Circular Layout")
```

Q1,Q2,Q5: Network Graph (Circular Layout)



Output — Q1, Q2 & Q5: Directed network graph in circular layout, colored nodes with labels

3. Temporal Data Representation

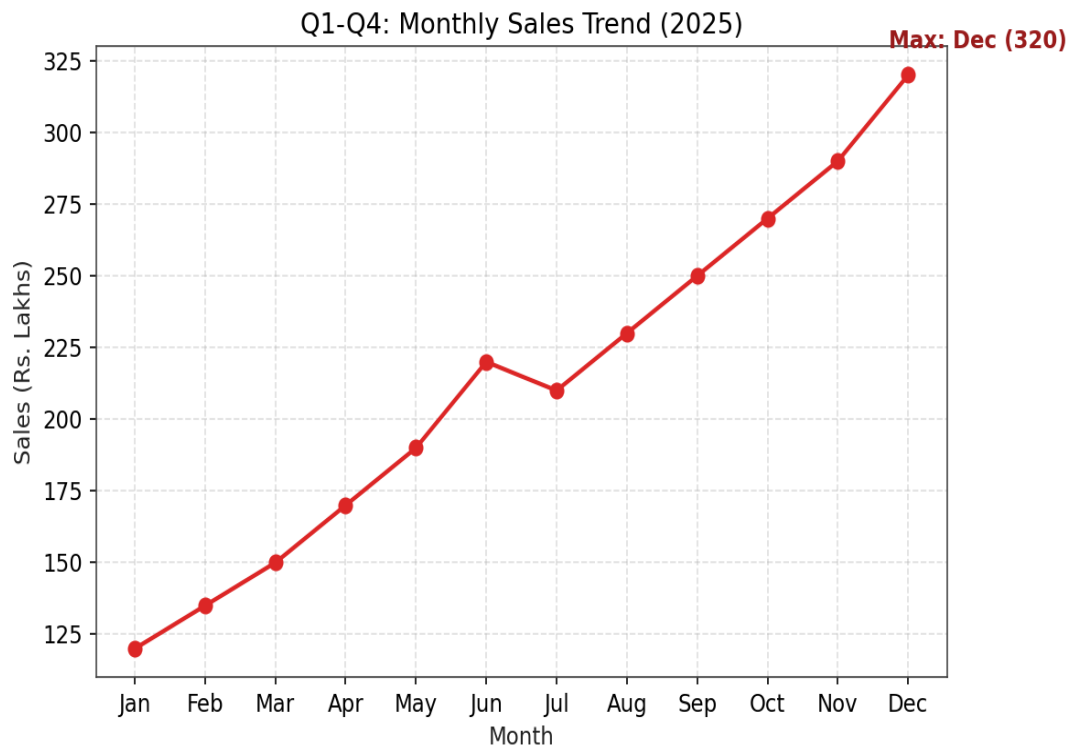
Dataset & Setup

```
sales <- c(120,135,150,170,190,220,210,230,250,270,290,320)
months <- c("Jan","Feb","Mar","Apr","May","Jun",
            "Jul","Aug","Sep","Oct","Nov","Dec")
```

Q1-Q4. Line chart with custom month axis, highlighted points, and grid lines

```
plot(1:12, sales, type = "o", col = "red", lwd = 2, pch = 19,
     xaxt = "n", xlab = "Month", ylab = "Sales (Rs. Lakhs)",
     main = "Monthly Sales Trend (2025)")

axis(1, at = 1:12, labels = months)      # Q2: custom month names
grid(nx = NA, ny = NULL, lty = "dashed", col = "gray") # Q4: grid lines
points(1:12, sales, pch = 19, col = "red") # Q3: highlighted points
```



Output — Q1-Q4: Monthly sales line chart with labeled axis, highlighted points, and grid

Q5. Month with maximum sales

Answer: December recorded the maximum sales at **Rs. 320 Lakhs**, continuing the overall upward trend through the year (with a small dip only in July, from 220 to 210).

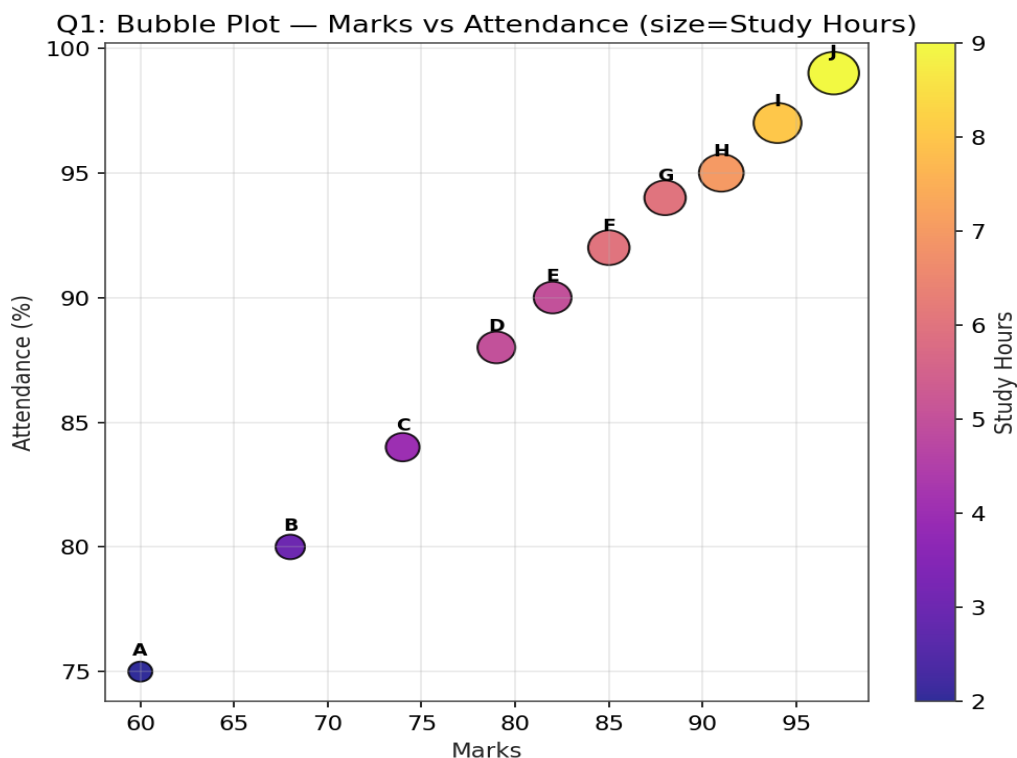
4. Multivariate Data Visualization

Dataset & Setup

```
students <- data.frame(  
  Student = c("A", "B", "C", "D", "E", "F", "G", "H", "I", "J"),  
  Marks = c(60, 68, 74, 79, 82, 85, 88, 91, 94, 97),  
  Attendance = c(75, 80, 84, 88, 90, 92, 94, 95, 97, 99),  
  StudyHours = c(2, 3, 4, 5, 5, 6, 6, 7, 8, 9),  
  Projects = c(1, 2, 2, 3, 3, 4, 4, 5, 5, 6)  
)
```

Q1. Bubble plot — Marks (X), Attendance (Y), Study Hours (bubble size)

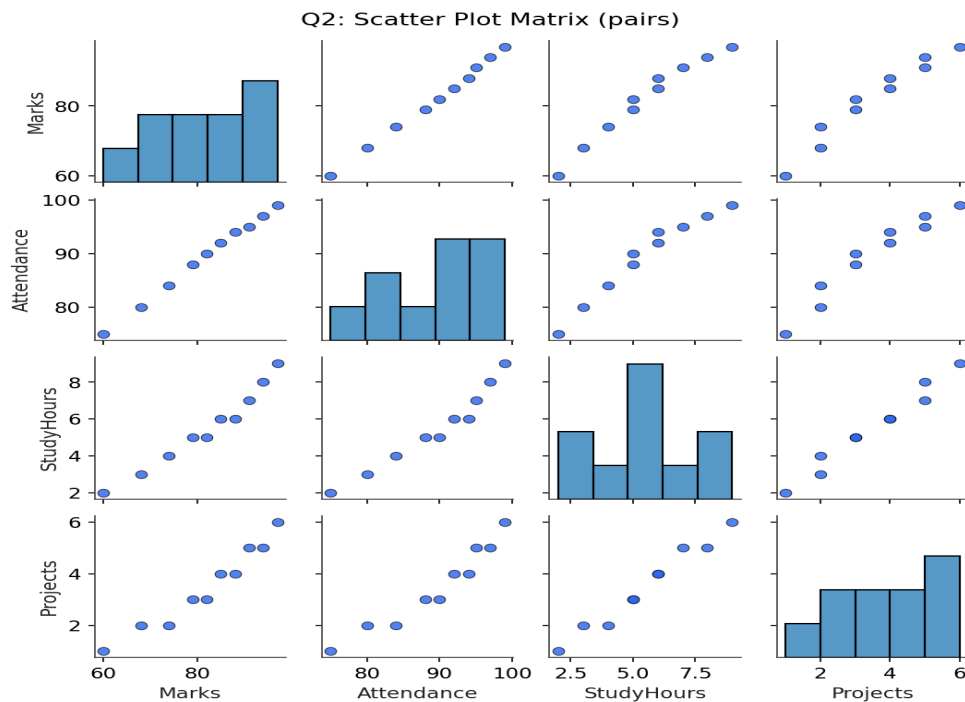
```
library(ggplot2)  
  
ggplot(students, aes(x = Marks, y = Attendance, size = StudyHours, color = StudyHours)) +  
  geom_point(alpha = 0.8) +  
  geom_text(aes(label = Student), vjust = -1.3, size = 3.5, color = "black") +  
  scale_size(range = c(3, 12)) +  
  labs(title = "Marks vs Attendance (Bubble Size = Study Hours)") +  
  theme_minimal()
```



Output — Q1: Bubble plot of Marks vs Attendance, sized by Study Hours

Q2. Scatter plot matrix using pairs()

```
pairs(students[, c("Marks", "Attendance", "StudyHours", "Projects")],  
      main = "Scatter Plot Matrix", col = "steelblue", pch = 19)
```



Output — Q2: Scatter plot matrix of numeric variables

Q3. Correlation among Marks, Attendance, and Study Hours

```
cor(students[, c("Marks", "Attendance", "StudyHours")])
```

	Marks	Attendance	StudyHours
Marks	1.000	0.998	0.981
Attendance	0.998	1.000	0.974
StudyHours	0.981	0.974	1.000

Interpretation: All three variables are very strongly and positively correlated ($r > 0.97$). Students who study more hours tend to attend more classes and score higher marks.

Q4. Student with the highest marks and largest bubble

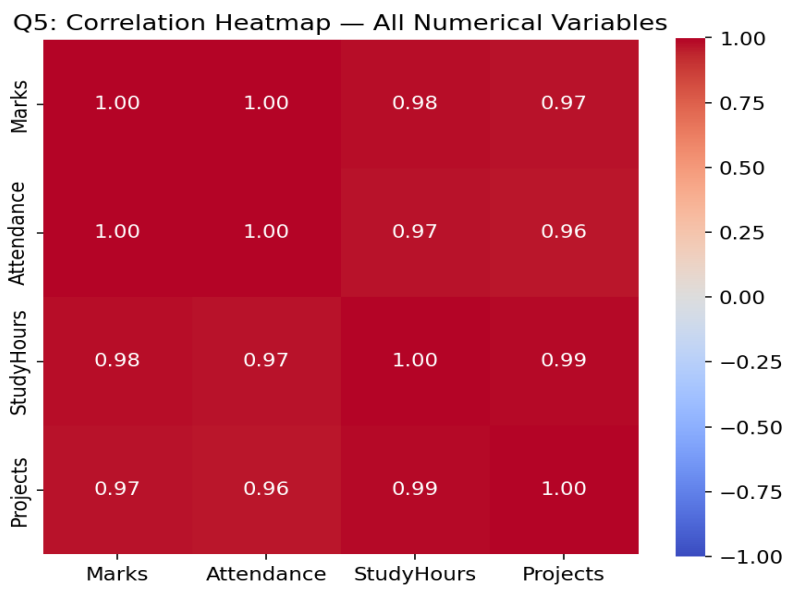
Answer: Student J has both the highest marks (97) and the largest bubble, reflecting the highest study hours (9) and attendance (99%) — visually confirming the strong positive relationship among these variables.

Q5. Heatmap of relationships among all numerical variables

```
library(reshape2)
library(ggplot2)

cor_matrix <- cor(students[, c("Marks", "Attendance", "StudyHours", "Projects")])
melted_cor <- melt(cor_matrix)

ggplot(melted_cor, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  geom_text(aes(label = round(value, 2)), color = "black", size = 3.5) +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red", midpoint = 0) +
  labs(title = "Correlation Heatmap", x = "", y = "") +
  theme_minimal()
```

Output — Q5: Correlation heatmap of Marks, Attendance, Study Hours, Projects

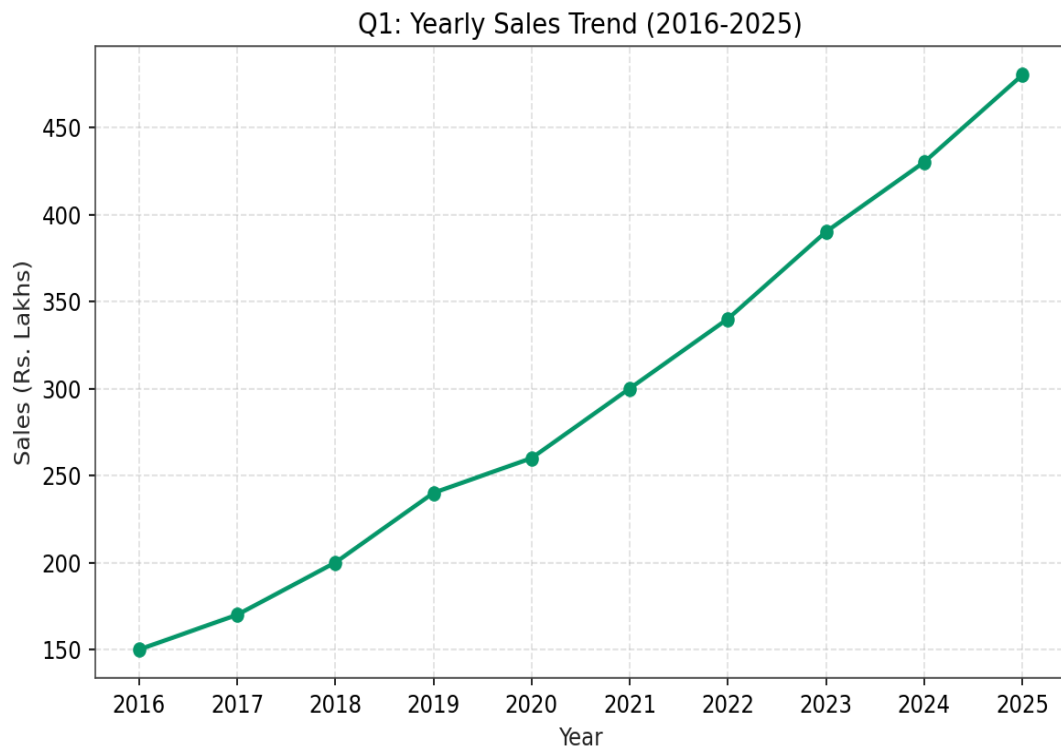
5. Integrated Data Visualization Exercise

Dataset & Setup

```
business <- data.frame(  
  Year = 2016:2025,  
  Sales = c(150,170,200,240,260,300,340,390,430,480),  
  Profit = c(20,25,32,40,42,55,65,72,82,95),  
  Customers = c(1200,1500,1800,2200,2400,2800,3300,3800,4200,4700),  
  Branches = c(5,6,7,8,8,10,11,12,13,15)  
)
```

Q1. Temporal line chart showing yearly sales

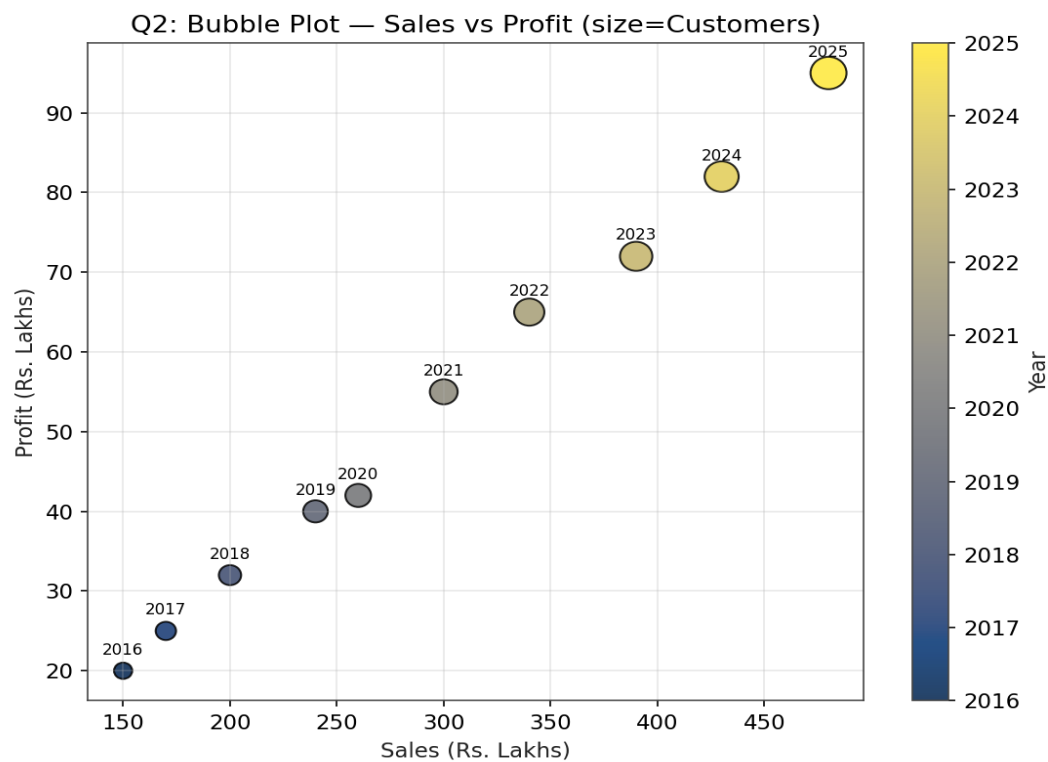
```
plot(business$Year, business$Sales, type = "o", col = "darkgreen", lwd = 2, pch = 19,  
      xlab = "Year", ylab = "Sales (Rs. Lakhs)",  
      main = "Yearly Sales Trend (2016-2025)")  
grid()
```



Output — Q1: Yearly sales line chart, 2016-2025

Q2. Multivariate bubble plot — Sales (X), Profit (Y), Customers (bubble size)

```
library(ggplot2)  
  
ggplot(business, aes(x = Sales, y = Profit, size = Customers, color = Year)) +  
  geom_point(alpha = 0.8) +  
  scale_size(range = c(3, 15)) +  
  labs(title = "Sales vs Profit (Bubble Size = Customers)") +  
  theme_minimal()
```



Output — Q2: Bubble plot of Sales vs Profit, sized by Customers, colored by Year

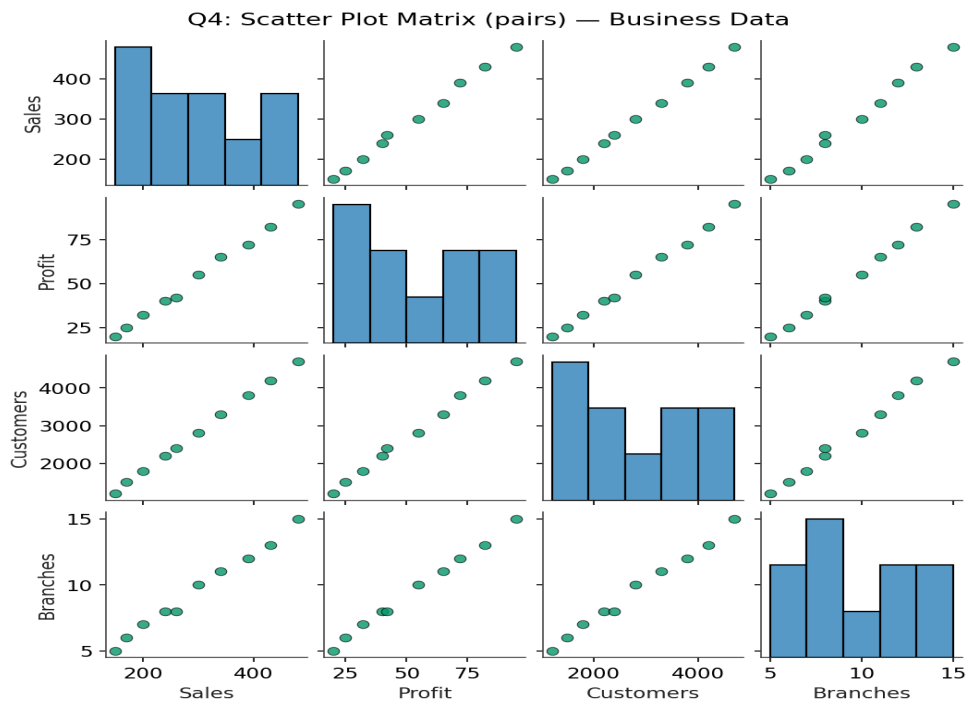
Q3. Correlation matrix for Sales, Profit, Customers, and Branches

```
cor(business[, c("Sales", "Profit", "Customers", "Branches")])
```

	Sales	Profit	Customers	Branches
Sales	1.000	0.998	1.000	0.996
Profit	0.998	1.000	0.998	0.999
Customers	1.000	0.998	1.000	0.996
Branches	0.996	0.999	0.996	1.000

Q4. Scatter plot matrix using pairs()

```
pairs(business[, c("Sales", "Profit", "Customers", "Branches")],
      main = "Scatter Plot Matrix - Business Data", col = "darkgreen", pch = 19)
```



Output — Q4: Scatter plot matrix of Sales, Profit, Customers, Branches

Q5. Interpretation: Does more branches associate with higher sales & profit?

Answer: Yes. Every visualization points to a strong positive relationship. The correlation matrix shows Branches correlates with Sales at $r = 0.996$ and with Profit at $r = 0.999$ — both near-perfect positive correlations. In the scatter plot matrix and line/bubble charts, Sales, Profit, Customers, and Branches all rise together in a consistent, near-linear pattern from 2016 to 2025 (e.g. Branches grew from 5 to 15 while Sales grew from Rs. 150L to Rs. 480L, more than tripling). This suggests that as the company opened more branches, it reached more customers, which in turn drove higher sales and profit — i.e., branch expansion appears to be a key growth driver, though correlation alone does not prove branches directly *caused* the growth (other factors like marketing or market demand could contribute as well).